

Drilling Mechanics - APWD
Time Based



Real Time, Recorded Mode

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude: 60° 34' 35.14" N

UWID:

Longitude: 3° 26' 36.13" E

Rig Name:

West Hercules

Block: 31/5

Rig Type:

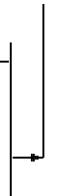
Semi-Submersible

FL: Norwegian North Sea

FL1: Section: 12 1/4 x 17 1/5 in.

FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level



Ground Level: 307.00 m

Acquisition Dates: 19-Dec-2019 -- 20-Dec-2019

Other Services:

Log Interval: 12/19/2019 10:00:00 AM -- 12/20/2019 5:00:00 PM

Direction & Inclination (Surveys)

Index Types: Time

Directional Drilling

Index Scales: 5 cm / 3600 secs

VISION Resistivity

Depth Source: Driller's Depth

Depth Sensor: 3rd Party Depth

Print Type: Final

Spud Date: 02-Dec-2019

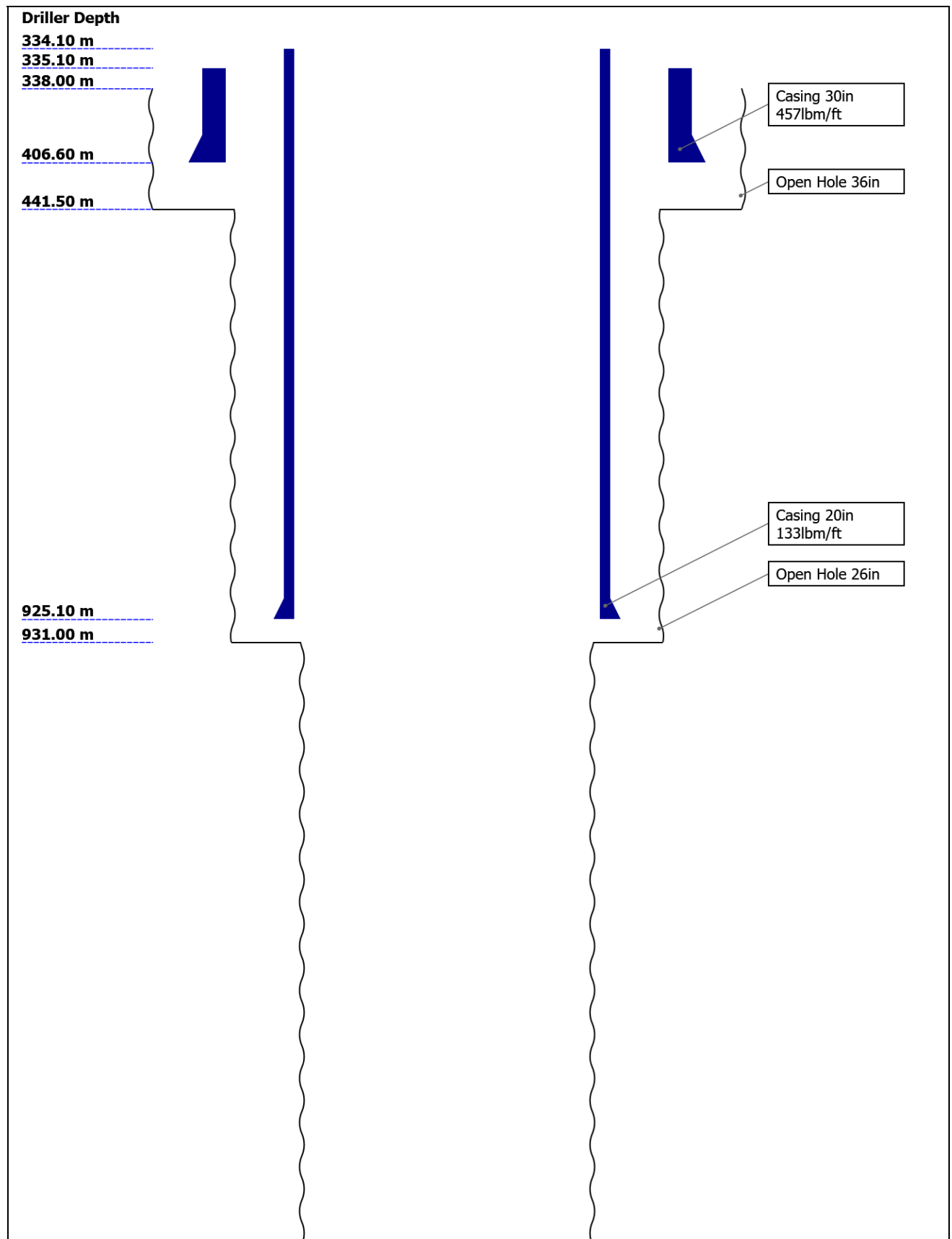
Disclaimer

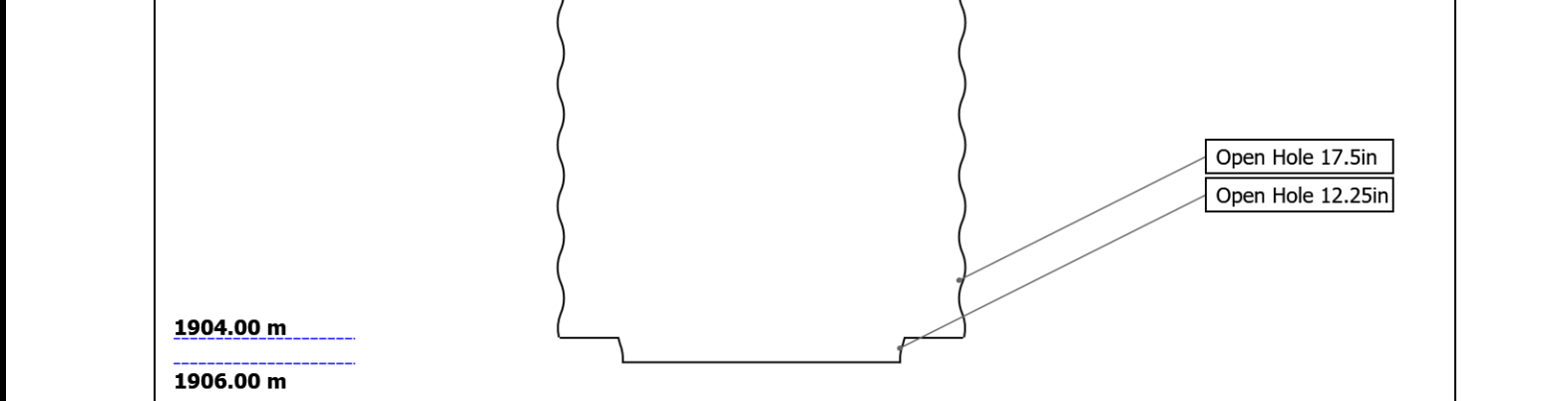
THE USE OF AND RELIANCE UPON THIS RECORDED-DATA BY THE HEREIN NAMED COMPANY (AND ANY OF ITS AFFILIATES, PARTNERS, REPRESENTATIVES, AGENTS, CONSULTANTS AND EMPLOYEES) IS SUBJECT TO THE TERMS AND CONDITIONS AGREED UPON BETWEEN SCHLUMBERGER AND THE COMPANY, INCLUDING: (a) RESTRICTIONS ON USE OF THE RECORDED-DATA; (b) DISCLAIMERS AND WAIVERS OF WARRANTIES AND REPRESENTATIONS REGARDING COMPANY'S USE AND RELIANCE UPON THE RECORDED-DATA; AND (c) CUSTOMER'S FULL AND SOLE RESPONSIBILITY FOR ANY INFERENCE DRAWN OR DECISION MADE IN CONNECTION WITH THE USE OF THIS RECORDED-DATA.

Contents

1. Header
2. Disclaimer
3. Contents
4. Well Sketch
5. Borehole Size/Casing/Tubing Record
6. Operational Run Summary
7. Borehole Fluids
8. Remarks and Equipment Summary
9. Run 7 Drilling Mechanics - APWD - Time Based
 - 9.1 Integration Summary
 - 9.2 Composite Summary
 - 9.3 Log (Drilling Mechanics Light)
10. Tail

Well Sketch





Borehole Size/Casing Record						
Bit						
Bit Size (in)	36	26	17.5	12.25		
Top Driller (m)	338	441.5	931	1904		
Bottom Driller (m)	441.5	931	1904	1906		
Casing						
Size (in)	30	20				
Weight (lbm/ft)	457	133				
Inner Diameter (in)	27.067	18.73				
Grade	X52	N80				
Top Driller (m)	335.1	334.1				
Bottom Driller (m)	406.6	925.1				

Operational Run Summary						
Parameter (unit)	Run 7					
Date Log Started	19-Dec-2019					
Time Log Started	05:51:07					
Date Log Finished	20-Dec-2019					
Time Log Finished	23:28:36					
Bit Size (in)	17.500					
Bit Start Depth (m)	931.00					
Bit Stop Depth (m)	1906.00					
Top Log Interval (m)	931.00					
Bottom Log Interval (m)	1894.18					
Max Hole Deviation (deg)	1.10					
Azimuth of Max Deviation (deg)	133.27					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	G.Boyd					
Witnessed By	S.Vikra					
Service Order Number	19SCA0085					

Borehole Fluids

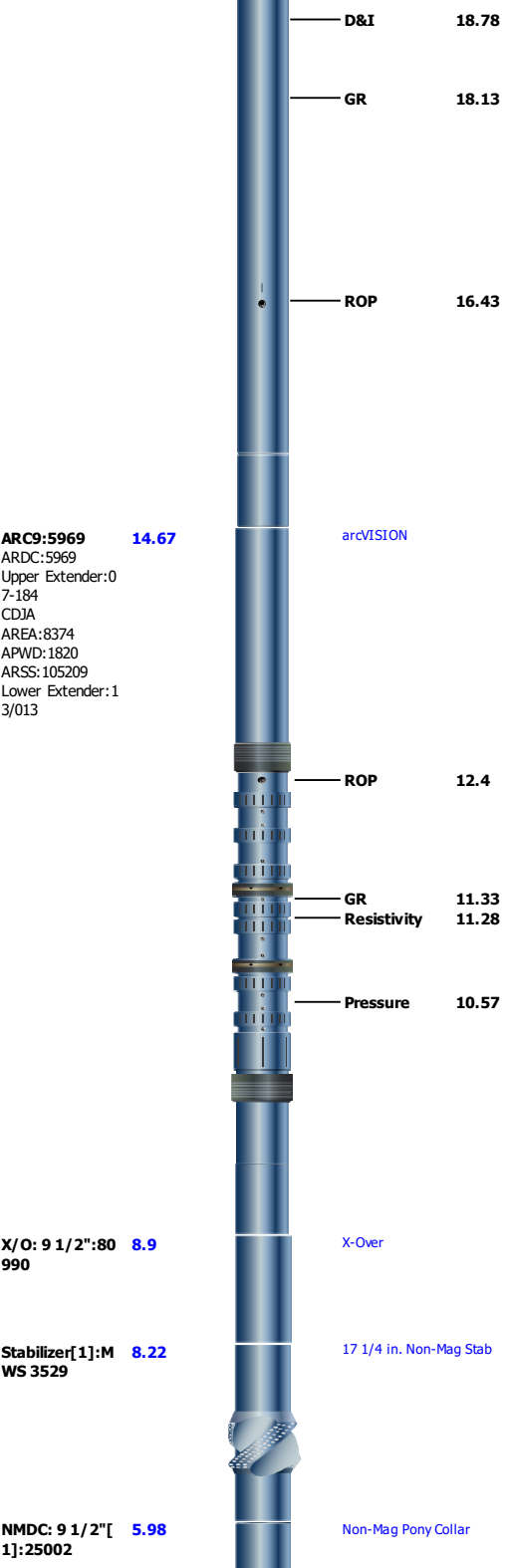
Parameter(unit)	Run 7					
Fluid Type	Water					
Fluid Name	Glydril					
Max Recorded Temperatures (degC)	38					
Source of Sample	Active Tank					
Salinity (ppm)	96819.84					
Density (g/cm3)	Zoned					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.09 @ 16.3					
RMF @ Meas Temp (ohm.m@degC)	0.06 @ 16					
RMC @ Meas Temp (ohm.m@degC)	0.12 @ 20					
RM @ BHT (ohm.m@degC)	0.06 @ 38					
RMF @ BHT (ohm.m@degC)	0.04 @ 38					
RMC @ BHT (ohm.m@degC)	0.08 @ 38					
Total Solid (%)						
High Gravity Solids (%)						

Run 7

Parameter	Value	Start
Density	1.25	12/19/2019 5:51:07 AM
Density	1.26	12/19/2019 11:26:00 AM

Remarks and Equipment Summary

Run 7: Toolstring			Run 7: Remarks
Equip name Stabilizer[2]:26 068	Length 28.34	MP name 17 1/4 in. Non-Mag Stab	All depths referenced to driller's depth. See EOWR for depth tracking details.
			APWD data is from tool memory. All other data is Real Time.
			Gamma Ray measurement is corrected for collar thickness only and not for Potassium content.
NMDC: 9 1/2"[2]:26105-1	26.00	Non-Mag Pony Collar	Run 7 and 12 1/4 x 17 1/2 in. section TD at 1906 m.
TELE900:G509 3 MSSU900:71539 Upper Extender:2 6-02-16 MDC900:G5093 MMA:2068 MDI:3956 PMGR:455 PME:3708 MTA:2526 MTK900:008/298 895-2142 MSS900:973247	23.35	TeleScope	



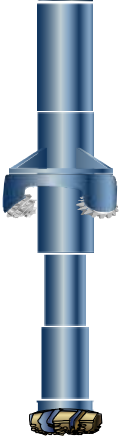
Fit Sub: 9 1/2" 4.00
:25003

Hole Opener:JK 3.34
4951

X/O: 6 3/4":IO 1.69
T 25162

X/O: 8 1/4":OW 1.28
S-BS-532

Bit: 12 1/4" 0.3



Float sub
17 1/2 in. Hole Opener
X-Over
X-Over
Smith PDC MDSIZ616
TOOL_ZERO

Lengths are in m
Maximum Outer Diameter = 17.500 in
Line: Sensor Location, Value: Gating Offset
All measurements are relative to TOOL_ZERO

Run 7

Drilling Mechanics - APWD - Time Based

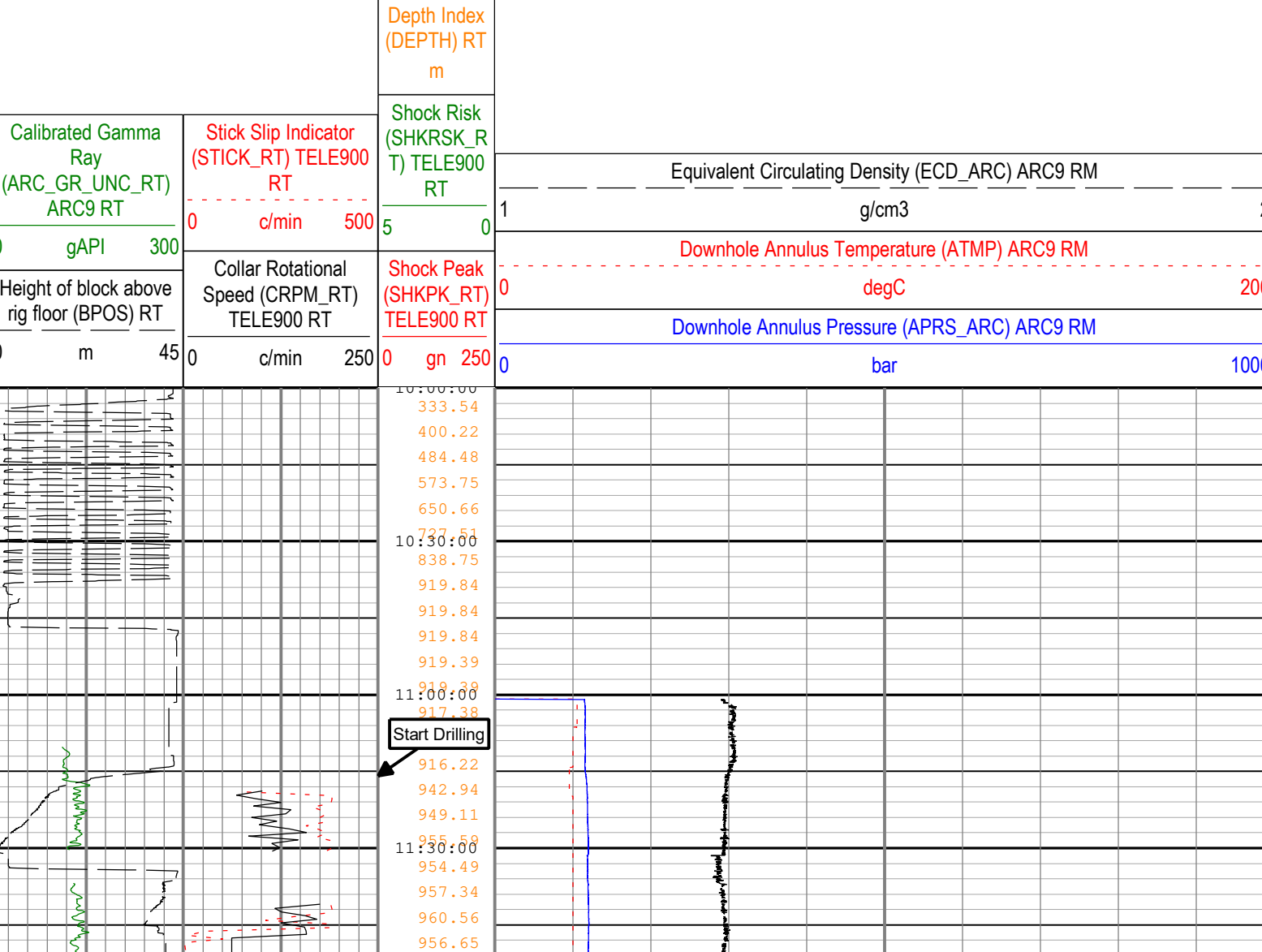
Pass Summary

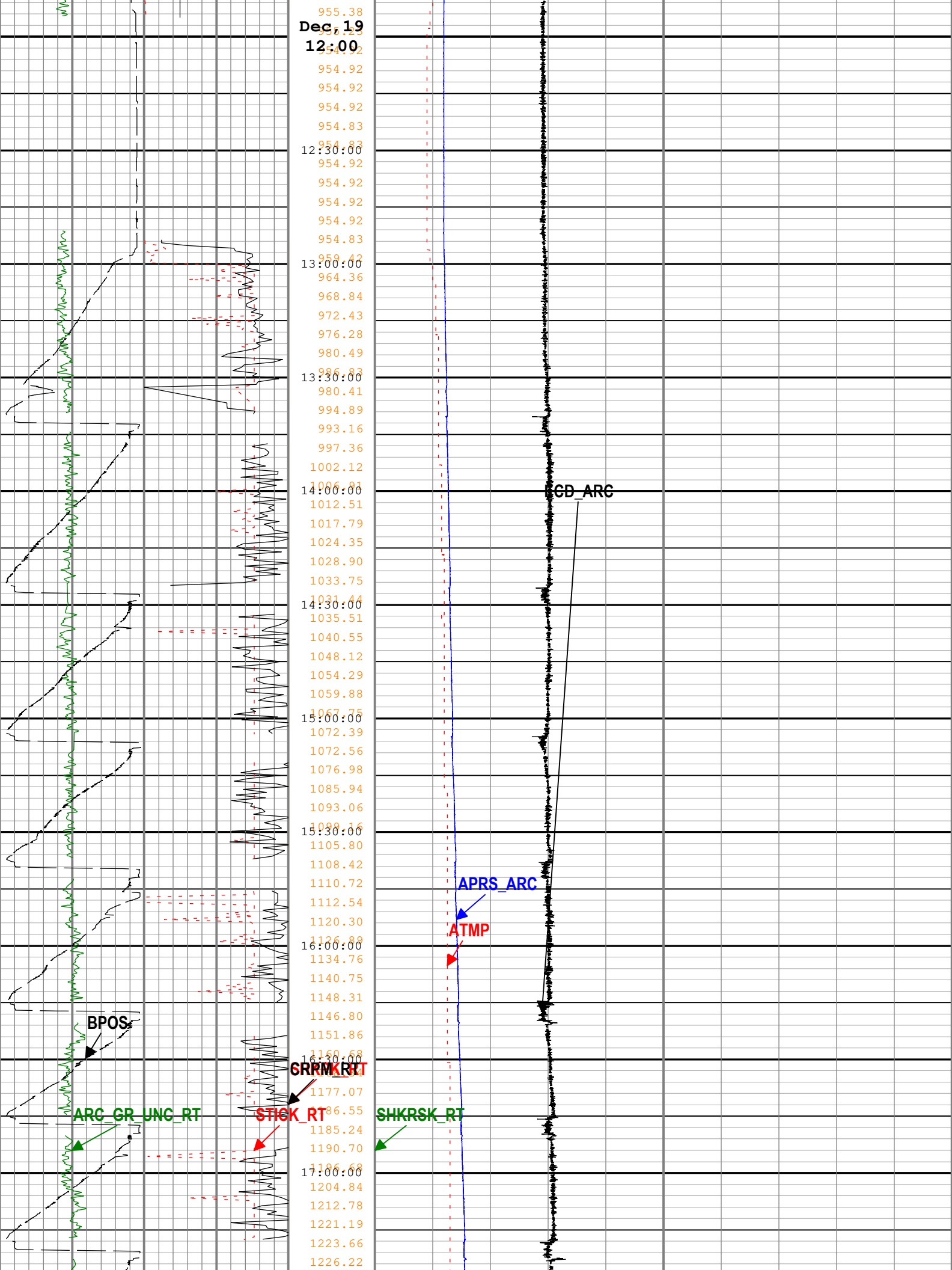
Run Name	Pass Objective	Direction	Start	Stop	Include Parallel Data
Run 7	TimeLogicalAcq	Down	19-Dec-2019 5:51:07 AM	20-Dec-2019 11:28:36 PM	Yes

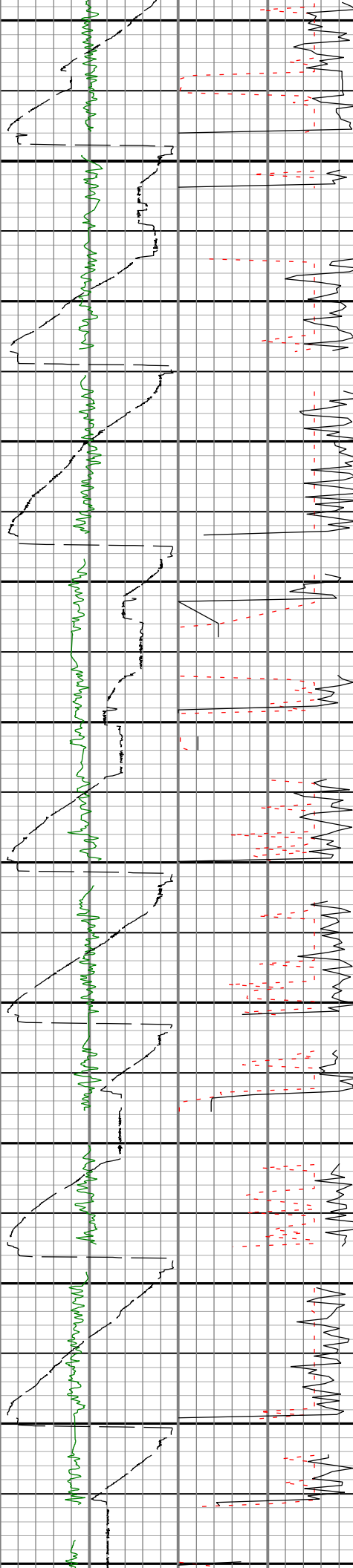
Log

Company:Equinor Well:31/5-7
Run 7: TimeLogicalAcq:S052

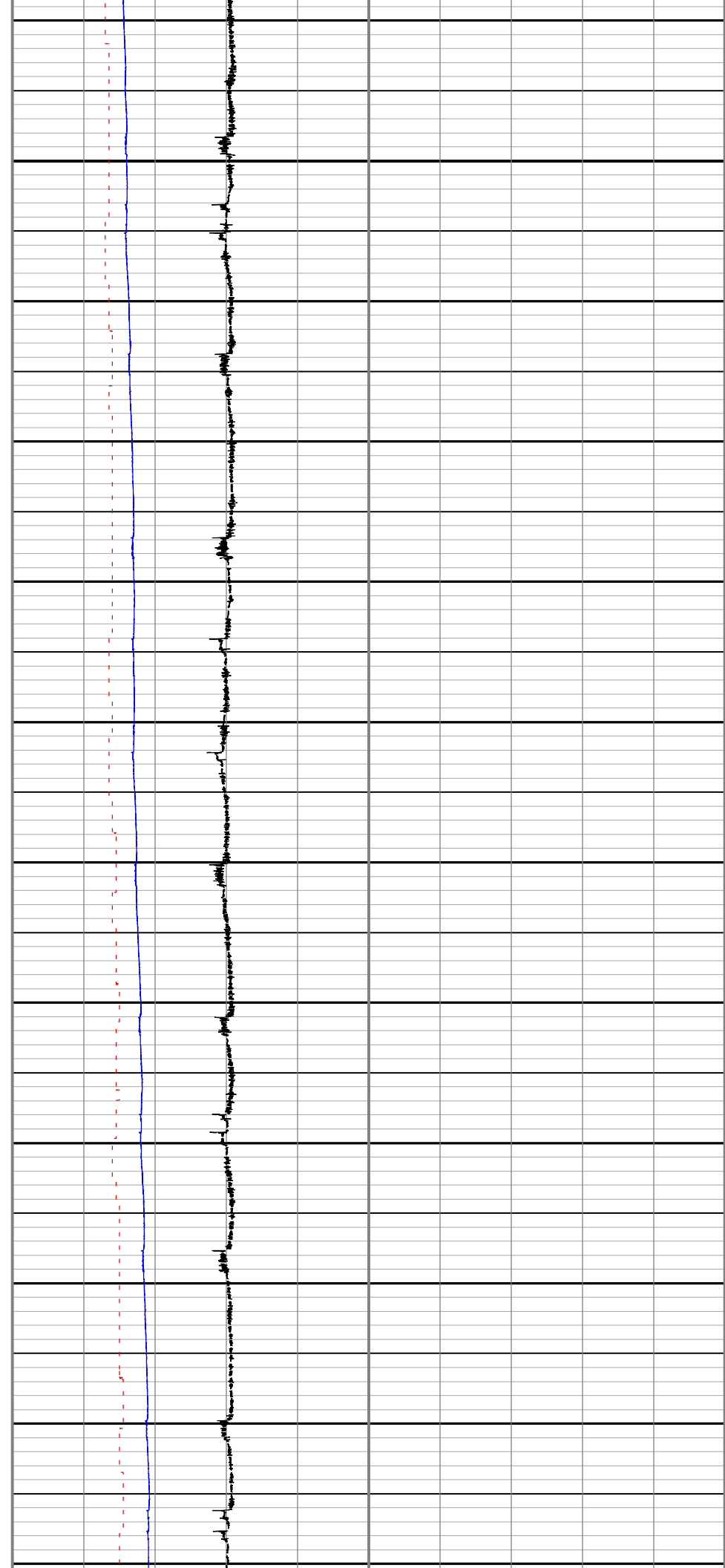
Description: Format: Log (Drilling Mechanics Light) Index Scale: 5 cm per 3600 s Index Type: Time Creation Date: 27-Dec-2019 15:44:05

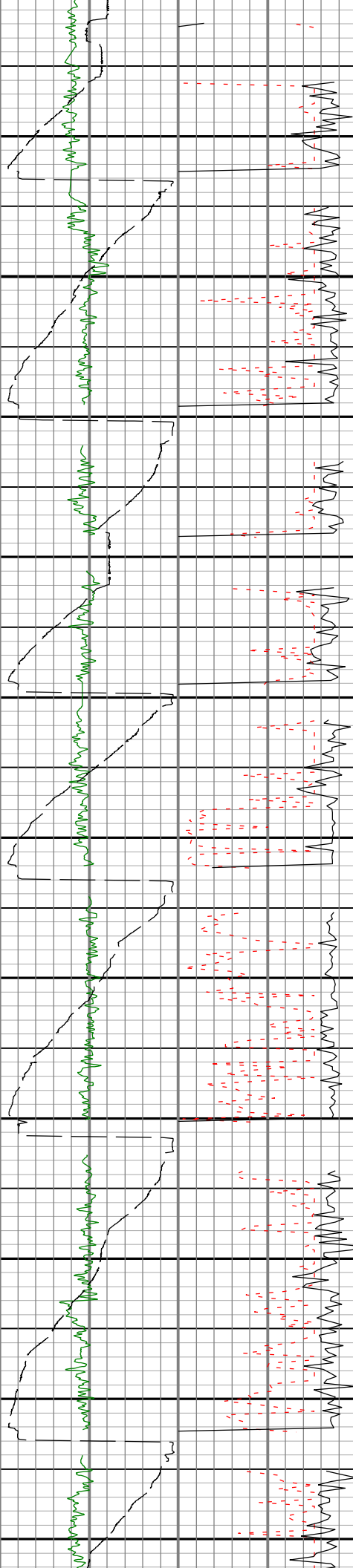




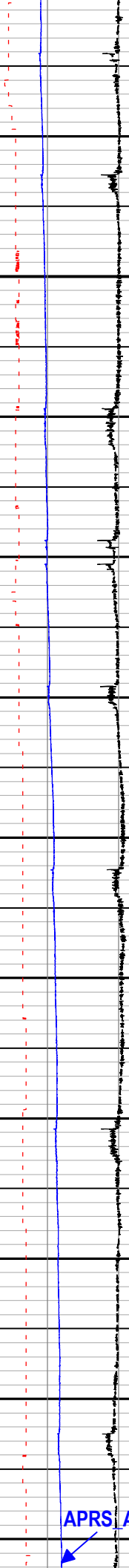


17:30:00
1238.64
1248.45
1248.81
1257.50
1264.62
Dec 19
18:00:00
1267.77
1267.72
1266.33
1274.55
1284.80
18:30:00
1291.78
1301.48
1300.69
1303.91
1310.84
1318.86
19:00:00
1325.33
1330.96
1336.76
1341.04
1339.27
1343.42
19:30:00
1347.80
1347.32
1346.94
1346.49
1353.59
1355.38
20:00:00
1351.55
1351.64
1358.20
1366.72
1374.36
1378.55
20:30:00
1377.80
1381.15
1389.16
1397.25
1406.59
1415.15
21:00:00
1416.20
1418.78
1425.90
1433.43
1428.77
1428.68
21:30:00
1429.57
1440.34
1449.49
1455.70
1454.35
1456.88
22:00:00
1463.41
1469.64
1477.87
1484.41
1491.09
1493.53
22:30:00
1495.25
1500.46
1508.33
1508.26
1508.16
1508.66
23:00:00

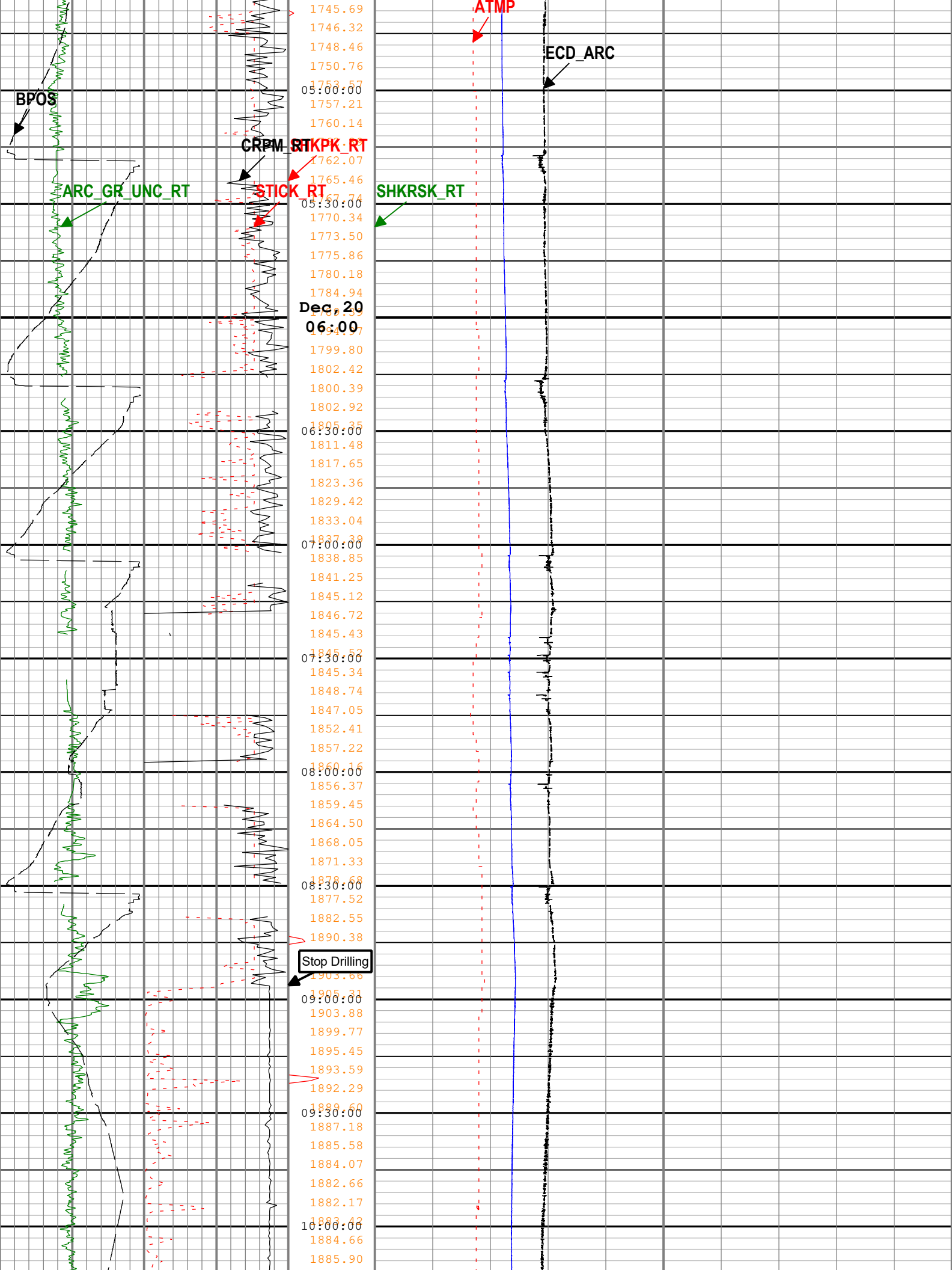


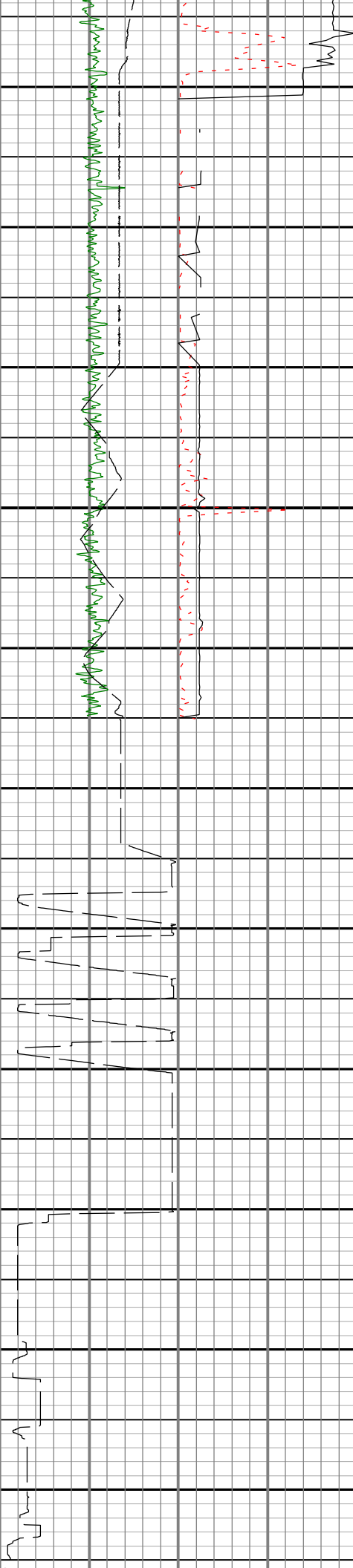


1508.35
1513.71
1509.88
1513.54
1518.86
23:30:00
1525.83
1530.50
1530.97
1534.03
1539.14
1545.79
Dec 20
00:00:00
1558.96
1559.13
1562.63
1567.99
1570.97
00:30:00
1588.11
1569.15
1572.41
1574.90
1580.28
1587.60
01:00:00
1585.05
1585.23
1590.23
1596.86
1603.39
1608.56
01:30:00
1607.89
1612.42
1617.71
1624.71
1632.66
1639.84
02:00:00
1644.16
1647.70
1646.49
1647.97
1653.37
1659.77
02:30:00
1662.76
1666.37
1672.14
1676.83
1681.64
1684.80
03:00:00
1687.19
1684.87
1686.23
1688.10
1693.11
1700.30
03:30:00
1702.30
1704.05
1708.83
1713.28
1718.48
1721.89
04:00:00
1723.82
1725.27
1723.48
1725.63
1728.27
1733.85
04:30:00
1738.41
1743.92

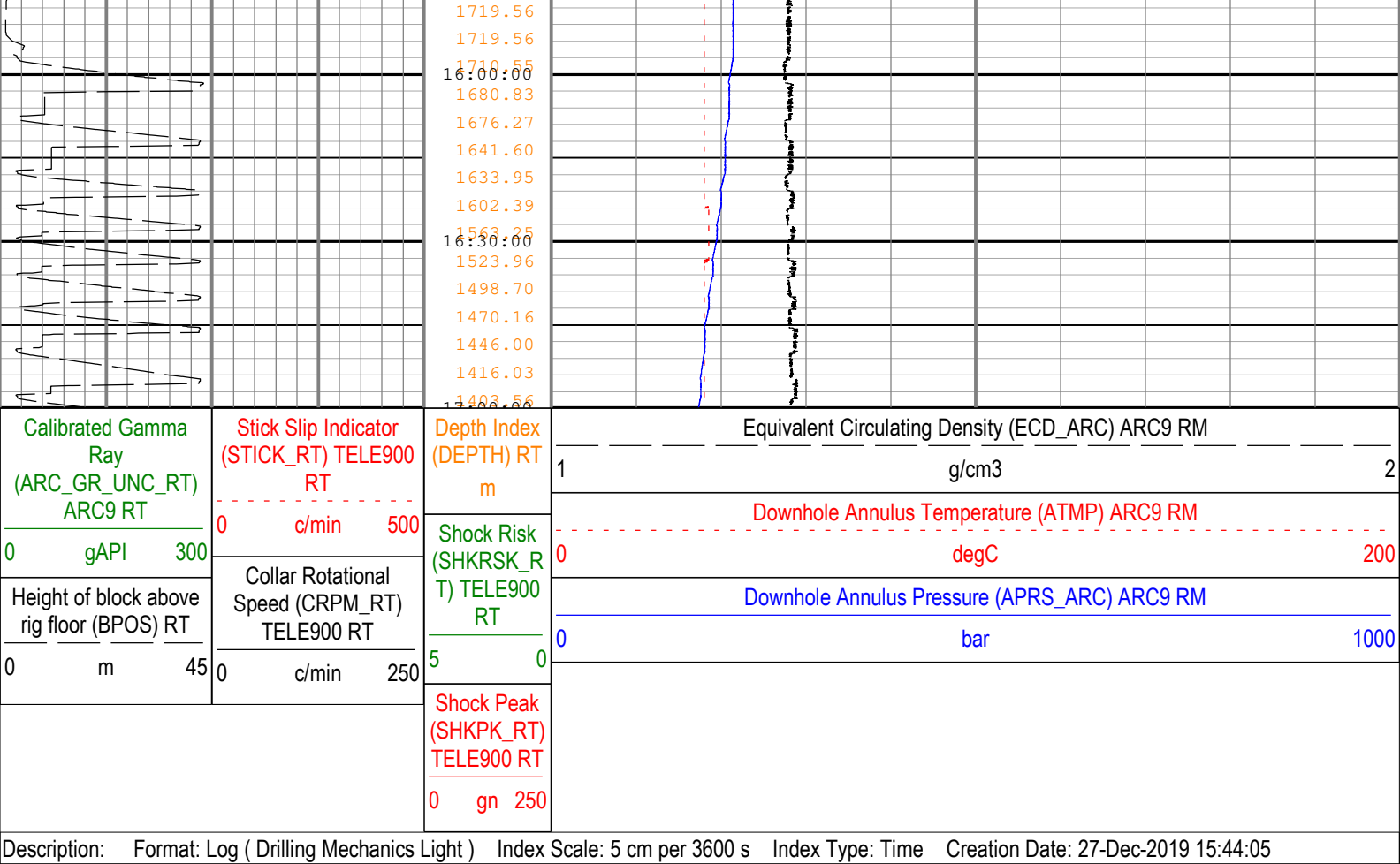


APRS ARC





1887.14
1888.27
1888.33
10:30:00
1890.21
1890.39
1890.30
1890.21
1890.21
1890.30
11:00:00
1890.39
1890.30
1890.30
1890.30
1890.21
11:30:00
1890.21
1894.49
1899.38
1896.11
1892.80
1889.77
Dec, 20
12:00
1897.05
1898.74
1895.01
1890.20
1892.54
12:30:00
1895.86
1899.09
1893.76
1891.37
1889.90
1889.90
13:00:00
1889.90
1889.90
1889.90
1883.42
1877.09
1877.09
13:30:00
1841.61
1837.79
1815.72
1798.63
1789.34
1759.35
14:00:00
1741.82
1719.93
1719.93
1719.93
1719.93
1719.93
14:30:00
1718.83
1719.99
1719.59
1719.59
1719.59
1719.59
15:00:00
1718.59
1723.19
1723.21
1723.21
1721.00
1719.63
15:30:00
1718.63
1719.56
1719.56
1719.56



Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway



Schlumberger

Drilling Mechanics - APWD
Time Based
Real Time, Recorded Mode